7.

$$\begin{array}{ccccccc} 1 & & -1 & & 3 & & 10 \\ & -2 & & 4 & & 7 & \\ & & 6 & & 3 & & \\ & & & -3 & & & \end{array}$$

h_n has n of degree 3, so the 0th diagonal sequence is

$1, -2, 6, -3, 0, 0, 0, \cdots$

then we can express h_n,

$$h_n = 1\cdot\binom{n}{0} + (-2)\binom{n}{1} + 6\binom{n}{2} + (-3)\binom{n}{3}$$

$$= 1 - 2\binom{n}{1} + 6\binom{n}{2} - 3\binom{n}{3}$$

$$= -\frac{1}{2}n^3 + \frac{9}{2}n^2 - 6n + 1 \quad //$$

$$\sum_{k=0}^{n} h_k = 1\cdot\sum_{k=0}^{n}\binom{k}{0} - 2\sum_{k=0}^{n}\binom{k}{1} + 6\sum_{k=0}^{n}\binom{k}{2} - 3\sum_{k=0}^{n}\binom{k}{3}.$$

From $\sum_{k=0}^{n}\binom{k}{p} = \binom{n+1}{p+1}$,

$$\sum_{k=0}^{n} h_k \Rightarrow 1\binom{n+1}{1} - 2\binom{n+1}{2} + 6\binom{n+1}{3} - 3\binom{n+1}{4}$$

#8. Find the sum of the fifth powers of the first n positive integers.

Sol) Let $h_n = n^5$

$$
\begin{array}{ccccccccc}
0 & 1 & 32 & 243 & 1024 & 3125 & & & \\
 & 1 & 31 & 211 & 781 & 2101 & & & \\
 & & 30 & 180 & 570 & 1320 & & & \cdots \\
 & & & 150 & 390 & 750 & & & \\
 & & & & 240 & 360 & & & \\
 & & & & & 120 & & &
\end{array}
$$

Because h_n is a polinomial of degree 5, the 0 th diagonal of the difference table equals

$0, 1, 30, 150, 240, 120, 0, 0, \ldots$

Hence

$$\sum_{k=0}^{n} k^5 = 0\binom{n+1}{1} + 1\binom{n+1}{2} + 30\binom{n+1}{3} + 150\binom{n+1}{4} + 240\binom{n+1}{5} + 120\binom{n+1}{6}$$

15. The number of partitions of a set of n elements into k distinguishable boxes (some of which may be empty) is k^n.
By counting in a different way, prove that

$$k^n = \binom{k}{1} 1!\, S(n,1) + \binom{k}{2} \cdot 2! \cdot S(n,2) + \cdots + \binom{k}{n} \cdot n! \cdot S(n,n).$$

(If $k > n$, define $S(n,k)$ to be 0).

pf) Consider the number of empty boxes. say n.

If $n = k-1$, only one box has n elements.

$$\Rightarrow \binom{k}{1} \cdot 1! \cdot S(n,1)$$

Choose only one box consisting n elements.

the number of arrangement of one box.

the number of partitions of a set of n elements into one box.
(i.e. The stirling number of the second kind)

If $n = k-2$, two boxes have n elements.

$$\Rightarrow \binom{k}{2} \cdot 2! \cdot S(n,2)$$

Choose two boxes consisting n elements

the number of arrangement of two boxes.

the number of partitions of a set of n elements into two boxes.

Continuing this process, the number of partition of a set of n element into k distinguishable boxes is

$$\binom{k}{1} \cdot 1! \cdot S(n,1) + \binom{k}{2} \cdot 2! \cdot S(n,2) + \cdots + \binom{k}{n} \cdot n! \cdot S(n,n).$$

By hypothesis, $k^n = \binom{k}{1} \cdot 1! \cdot S(n,1) + \cdots + \binom{k}{n} \cdot n! \cdot S(n,n)$.

$\blacksquare$

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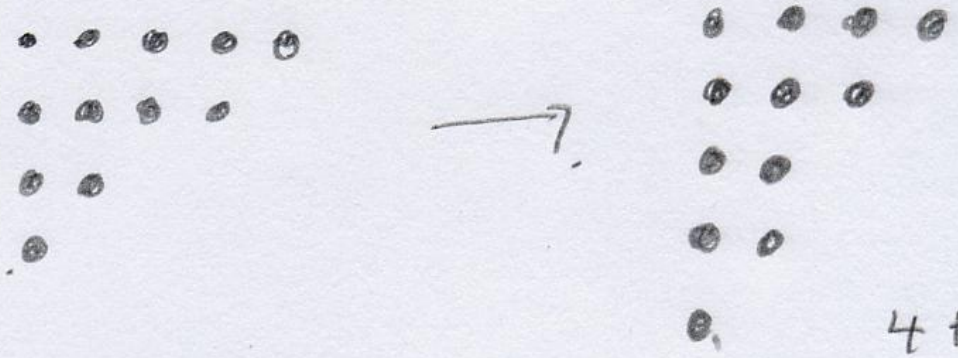
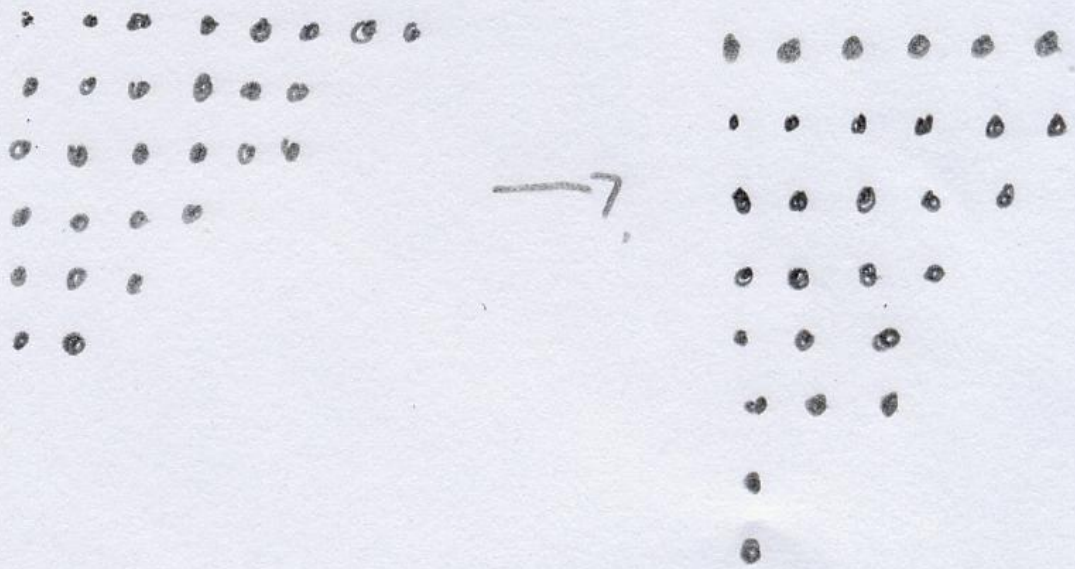
(a). $12 = 5 + 4 + 2 + 1$



$4 + 3 + 2 + 2 + 1$

(e) $29 = 8 + 6 + 6 + 4 + 3 + 2$



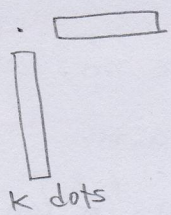
$6 + 6 + 5 + 4 + 3 + 3 + 1 + 1$

6.27 For each integer $n>2$, determine a self-conjugate partition of n that has at least two parts.

sol) If n is odd, put $n=2k+1$ $(k\in\mathbb{N})$.

Then a self conjugate of partition of n is the form of

that is



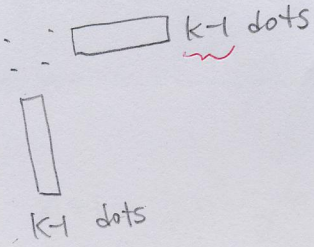
So, we can make self conjugate of odd n.

If n is even, put $n=2k$ $(k\in\mathbb{N}-\{1\})$

Then to make a self conjugate of partition of n,

first draw 4 dots

Then draw



like this.

So, we can make self conjugate of even n.

If n is odd, then

$\frac{n+1}{2}+1+\dots+1$ is a self-conjugate partition.

If n is even then

$\frac{n}{2}+2+1+\dots+1$ is a self conjugate partition

39 For every partition

$$a_1 + a_2 + \dots + a_r$$

of $n-1$,

$$a_1 + a_2 + \dots + a_r + 1$$

is a partition of n

So $p_n \geq p_{n-1}$.

The trivial partition n, of n isn't constructed in this way, so

$$p_n > p_{n-1}.$$